Dependency Parsing Across the Resource Spectrum: Evaluating Architectures on High and Low-Resource Languages

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.

Abstract

Transformer-based models achieve state-of-the-art dependency parsing for high-resource languages, yet their advantage over simpler architectures in low-resource settings remains poorly understood. We evaluate four parsers—the Biaffine LSTM, Stack-Pointer Network, AfroXLMR-large, and RemBERT—across ten typologically diverse languages, with a focus on low-resource African languages. We find that the Biaffine LSTM consistently outperforms transformer models in low-resource regimes, with transformers recovering their advantage as training data increases. The crossover falls within a resource range typical of treebanks for under-resourced languages. Morphological complexity (measured via MATTR) emerges as a significant secondary predictor of transformers' relative disadvantage after controlling for corpus size. These results indicate that the Biaffine LSTM may be better suited for syntactic tool development in low-resource regimes until sufficient annotated data is available to leverage the representational capacity of pre-trained transformers.

1 Introduction

Dependency parsing—the task of identifying grammatical relations between words in a sentence—is a foundational task in NLP with broad downstream applications, including machine translation, information extraction, and semantic role labeling. Transformer-based models such as BERT and its multilingual variants now dominate dependency parsing benchmarks for high-resource languages, where abundant pre-training corpora and large treebanks allow these parameter-rich architectures to perform extremely well (Kondratyuk and Straka, 2019; Li et al., 2019; Grünewald et al., 2021).

This advantage, however, does not transfer uniformly to low-resource languages. Many languages—particularly African languages such as Xhosa and Wolof, spoken by millions but severely underrepresented in large text corpora—have dependency treebanks containing only a few hundred to a few thousand sentences, compared to tens of thousands available for English (Jha et al., 2020; Zhou et al., 2025). Pre-trained transformers require substantial in-domain data to fine-tune effectively, and the computational demands of fine-tuning present additional barriers in low-resource settings (Wang et al., 2020; Wu and Dredze, 2020).

We hypothesize that traditional architectures—particularly LSTM-based parsers—remain competitive with or outperform transformers when training data is scarce, for three reasons. First, **generalization**: transformer models contain orders of magnitude more learnable parameters than LSTM-based parsers. With limited training data, smaller models may generalize more reliably and overfit less (Gessler and Zeldes, 2022; Lindenmaier et al., 2025). Second, **inductive bias**: LSTM networks encode stronger built-in assumptions about sequential and hierarchical structure, which may serve as useful guidance when data is scarce (Tran et al., 2018; Dehghani et al., 2018). Third, **stability**: LSTM-based models tend to exhibit more consistent behavior across training runs, a meaningful advantage when working with small, noisy datasets (McCoy et al., 2020; Dodge et al., 2020).

To test this hypothesis, we systematically evaluate two LSTM-based parsers (Biaffine and Stack-Pointer) and two transformer-based parsers (AfroXLMR-large and RemBERT) across a diverse set of languages spanning a wide range of treebank sizes. Our central questions are: (1) Do LSTM-based parsers outperform transformer-based parsers on low-resource languages, while transformers regain their advantage at higher re-

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source levels? (2) Does the transformers' relative error disadvantage decrease monotonically as training data grows? We measure performance using labeled and unlabeled attachment scores (LAS/UAS), and we operationalize the transformer's relative advantage as its change in error rate normalized against the Biaffine LSTM baseline—a measure we call relative error rate (RER). This measure accounts for the fact that LAS gains are not comparable across languages and settings with very different baseline difficulties.

2 Related Work

Most African languages remain severely under-resourced in NLP due to their near-absence from web-crawled corpora such as Wikipedia and CommonCrawl, producing a digital language divide in which standard NLP pipelines fail to generalize to African linguistic contexts (Adelani, 2025; Bella et al., 2023; Nekoto et al., 2020).

Recent efforts to close this gap have proceeded along two complementary tracks: dataset construction and model adaptation. In the data domain, various initiatives have produced word embeddings (Adelani, 2022; Bojanowski et al., 2017), human-curated parallel corpora, and labeled benchmarks for machine translation (Nsumba et al., 2026; Kalejaiye et al., 2025), sentiment analysis (Muhammad et al., 2023), and news classification (Adelani et al., 2023) across a wide range of languages. On the architectural front, AfriBERTa (Ogueji, 2022) and AfroXLMR (Alabi et al., 2022) adapt multilingual transformer encoders to African morphology and orthography. AfriBERTa is trained from scratch, while AfroXLMR leverages continued pre-training on African-language corpora; both outperform general baselines such as mBERT and XLM-R on African NLP tasks. Masakhane, a grassroots research community, has been central to both efforts, providing named-entity and part-of-speech benchmarks across 20 languages through MasakhaNER (Adelani et al., 2022) and MasakhaPOS (Dione et al., 2023), among other contributions.

3 Data

Syntactic resources for African languages have developed more slowly than the broader benchmarks described above. Treebanks such as Wolof-WTB (Dione, 2019) and Naija-NSC (Caron et al., 2019) establish gold-standard dependency annotations for individual languages, but broad cross-lingual coverage has remained limited. AfriSUD (Buzaaba et al., 2026) represents a significant step forward, providing human-annotated treebanks in the Surface-Syntactic Universal Dependencies (SUD) framework across nine African languages. This collection forms the basis of the present study, with seven of the nine languages—Swahili, Kinyarwanda, Nigerian Pidgin, Wolof, Yoruba, Xhosa, and Hausa—included in our evaluation.

Languages and resource groupings. While the AfriSUD collection provides data for nine languages, we exclude Efik and Runyankore from our evaluation because pre-trained static word embeddings were unavailable. To span a broader resource spectrum and establish high-resource baselines, we supplement the remaining seven AfriSUD languages with three additional SUD treebanks: French, Romanian, and Afrikaans. This yields a final evaluation set of ten languages: Swahili (swa), Kinyarwanda (kin), Nigerian Pidgin (pcm), Wolof (wol), Yoruba (yor), Hausa (hau), Xhosa (xho), French (fra), Romanian (ron), and Afrikaans (afr).

Treebank sources and splits. Our treebanks are SUD-converted versions of standard UD releases: French GSD (Abeillé et al., 2019), Romanian RRT (Barbu Mititelu and Irimia, 2016), AfriBooms for Afrikaans (Augustinus et al., 2016), and the AfriSUD collection (Buzaaba et al., 2026) for remaining languages. Splits follow an 80–10–10 ratio. Because the French training set is much larger than needed to characterize high-resource behavior, we randomly subsample it to 10,000 sentences; dev and test sets remain unchanged. Sentence counts after subsampling are reported in Table 1.

Language	Code	Train	Dev	Test
French	fra	10,000*	1,634	1,634
Romanian	ron	7,619	953	952
Afrikaans	afr	1,547	194	193
Swahili	swa	1,187	149	148
Kinyarwanda	kin	1,098	137	137
Nigerian Pidgin	pcm	1,085	135	136
Wolof	wol	736	92	92
Yoruba	yor	694	87	87
Hausa	hau	690	86	86
Xhosa	xho	369	46	46

Table 1: Treebank splits (sentence counts). *French training set subsampled from 13,074 to 10,000; dev/test sets unchanged.

Morphological complexity. To examine whether typological properties of the target language modulate architecture-level differences in parsing performance, we include morphological complexity as a covariate in our statistical models. We operationalize complexity via the Moving-Average Type–Token Ratio (MATTR) (Covington and McFall, 2010), which corrects the well-known sensitivity of raw type–token ratio to text length by averaging type–token ratios over a sliding window. Languages with richer morphology produce more distinct surface forms per window, yielding higher MATTR values. We use the gold tokenization provided by our treebanks and a window size of 500 tokens. MATTR scores are computed over each language's training split and reported in Table 2.

The resulting rankings align with established characterizations of morphological complexity. Xhosa and Kinyarwanda—both Bantu languages with productive agglutinative morphology (Probert, 2019; Nzeyimana, 2024)—rank first and second, reflecting the large inventory of distinct surface forms generated by concatenative affixation. Romanian ranks third, consistent with its retention of nominal case morphology distinguishing it from its Romance relatives (Maiden et al., 2021). At the other extreme, Nigerian Pidgin and Afrikaans score lowest, in line with the morphological reduction characteristic of pidginization and Afrikaans's well-documented morphological simplification from Dutch (McWhorter, 2001; Donaldson, 1993).

Language	Tokens	MATTR	z-score
Xhosa	6,282	0.800	+2.381
Kinyarwanda	26,700	0.673	+0.789
Romanian	173,946	0.634	+0.302
Swahili	29,781	0.626	+0.203
Yoruba	10,422	0.586	−0.285
French	244,944	0.570	−0.494
Wolof	15,415	0.567	−0.528
Hausa	13,874	0.566	−0.539
Nigerian Pidgin	26,489	0.541	−0.853
Afrikaans	39,199	0.531	−0.975

Table 2: MATTR scores and standardized values, computed over training splits with window size $w = 500$.

4 Models

We evaluate three primary dependency parsing architectures: the Biaffine LSTM parser (Dozat and Manning, 2017), transformer-based biaffine parsers, and the Stack-Pointer Network (Ma et al., 2018). The biaffine parsers are implemented using the SuPar Python package (Zhang, 2024), and the Stack-Pointer parsers are implemented using the NeuroNLP2 repository (Ma, 2024).

Biaffine LSTM. The deep biaffine attention parser (Dozat and Manning, 2017) is our graph-based baseline and the de facto standard for neural dependency parsing. It encodes sentences with a BiLSTM and scores arcs via a biaffine function, decoding the optimal tree with the Chu-Liu/Edmonds algorithm (Chu, 1965; Edmonds, 1967). We use this model with static FastText embeddings.

Stack-Pointer Network. The Stack-Pointer Network (Ma et al., 2018) is a transition-based parser with a BiLSTM-CNN encoder that decodes top-down and depth-first via a pointer network, incorporating ancestor and sibling features for higher-order context. This approach avoids the directional bias of traditional transition parsers while remaining more efficient than MST-based decoding.

AfroXLMR-large. AfroXLMR-large (Alabi et al., 2022) is a domain-adapted variant of XLM-R (Conneau et al., 2020) that continues pre-training on 17 African languages, yielding richer representations for the morphologically complex languages in our evaluation set. It is paired with SuPar's biaffine parsing head and fine-tuned end-to-end.

RemBERT. RemBERT (Chung et al., 2020) is a multilingual transformer encoder that decouples input and output embeddings, allocating a larger input layer to better handle morphologically rich languages. Like AfroXLMR-large, it is paired with a biaffine head in SuPar and fine-tuned jointly during training.

5 Experiments

Hyperparameter optimization. We tune each model via grid search with a fixed seed. To keep the search tractable, high-resource languages are downsampled to 5,000 training sentences, and models are trained at 25% of the full epoch budget with halved early-stopping patience. For the Biaffine LSTM we search over learning rate, decay rate, and decay steps (12 runs per language); for the transformer-based models (AfroXLMR-large and RemBERT) we search over learning rate, warmup steps, and the ratio between the encoder and task-head learning rates (18 runs); and for Stack-Pointer we additionally tune unknown-word replacement probability and gradient clipping (24 runs). All other hyperparameters are kept at default values as defined by the respective implementation libraries.

Word embeddings. The Biaffine LSTM and Stack-Pointer models use 300-dimensional pre-trained static word vectors. For languages in the official FastText distribution (Grave et al., 2018) we use vectors trained on CommonCrawl and Wikipedia. For African languages not covered there, we fall back to embeddings provided by Adelani's AfricaNLP resources repository (Adelani, 2022). As noted in Section 3, Efik and Runyankore are excluded from our analysis because no pre-trained vectors could be located for either language. Static vectors are concatenated with character-level representations (CNN or LSTM) and gold POS tag embeddings. The transformer-based models do not use static vectors; they generate contextualized subword representations internally.

Training protocol. Final models are trained on the full data with the best hyperparameters. Biaffine models run for up to 1,000 epochs (patience 100); Stack-Pointer models run for up to 600 epochs (patience 60). In practice, no model approached its epoch ceiling before early stopping triggered. Transformer models on high-resource languages (Romanian and French) are additionally capped at 100 epochs to balance computational costs, as these models require fewer passes to converge due to their large volumes of data. Each configuration is trained five times with different random seeds, and we report mean UAS and LAS across runs to reduce sensitivity to initialization.

6 Results

6.1 Parsing Performance

Tables 3a and 3b report mean LAS and UAS with standard deviations across five training runs for all four architectures and ten languages. Overall, performance scales with training set size, though the relative ordering of architectures varies by language. The transformer models outperform the Biaffine LSTM on higher-resource languages, while the LSTM is competitive with or superior to both transformers on several lower-resource ones. The Stack-Pointer parser is the weakest architecture across nearly all languages and metrics.

We quantify architecture-level differences using the *relative error rate* (RER). Let $\text{LAS}_{\text{Biaffine LSTM}}(\ell)$ and $\text{LAS}_{\text{TF}}(\ell)$ denote the labeled attachment scores of the Biaffine LSTM baseline and a transformer comparison model on language ℓ. We define:

$$RER_{\text{LAS}}(\ell) = \frac{\text{LAS}_{\text{Biaffine LSTM}}(\ell) - \text{LAS}_{\text{TF}}(\ell)}{100 - \text{LAS}_{\text{Biaffine LSTM}}(\ell)} \tag{1}$$

and likewise for unlabeled attachment (RER_{UAS}). We prefer RER over raw score differences for two reasons. First, absolute LAS or UAS gains are confounded with baseline difficulty: a 2-point gain at a baseline of 95 is not comparable to the same gain at a baseline of 70, as the remaining headroom differs substantially. Second, because attachment scores are bounded at 100, raw differences are heteroscedastic across the training-size axis, with low-resource languages exhibiting lower baselines and higher variance; normalizing against the remaining error stabilizes variance across languages. A value $RER_{\text{LAS}} > 0$ indicates that the comparison model incurs more errors than the Biaffine LSTM, while $RER_{\text{LAS}} < 0$ indicates fewer errors.

Tables 4 and 5 report RER_{LAS} and RER_{UAS} for each comparison model. Both transformer models show $RER_{\text{LAS}}, RER_{\text{UAS}} < 0$ (fewer errors than the LSTM) on French, Romanian, and Afrikaans, but $RER > 0$ on Xhosa, Wolof, and—for RemBERT—Swahili. Looking only at RER_{LAS}, the pattern of LSTM's competitiveness extends further. AfroXLMR shows positive RER_{LAS} on Swahili ($+0.049$) in addition to Xhosa ($+0.148$) and Wolof ($+0.077$). RemBERT's label-level underperformance is even more extensive: beyond Swahili ($+0.181$), Wolof ($+0.130$), and Xhosa ($+0.314$), it records positive RER_{LAS} on Yoruba ($+0.071$) and Kinyarwanda ($+0.042$). The Stack-Pointer parser consistently underperforms, with $RER > 0$ on nearly every language. Figure 1 plots RER_{LAS} and RER_{UAS} for each non-baseline model across languages ordered by training set size.

6.2 Training Data Scaling

To test whether RER is systematically related to training set size, we fit linear mixed models predicting RER_{LAS} and RER_{UAS} as a function of log training sentences, with language as a random intercept. We restrict this analysis to the two transformer models, since the Stack-Pointer parser is consistently outperformed by the LSTM baseline, showing $RER > 0$ on nearly every language.

Both models show a statistically significant negative relationship between training size and RER—RER decreases as more data becomes available.

Language	Biaffine LSTM	AfroXLMR-large	RemBERT	Stack-Pointer
French (fra)	93.51 ± 0.13	95.81 ± 0.05	95.53 ± 0.08	92.41 ± 0.05
Romanian (ron)	87.23 ± 0.18	90.61 ± 0.12	90.51 ± 0.06	87.41 ± 0.12
Afrikaans (afr)	85.85 ± 0.31	88.90 ± 0.28	87.75 ± 0.19	85.91 ± 0.61
Swahili (swa)	79.48 ± 0.43	78.48 ± 0.49	75.77 ± 0.53	73.10 ± 0.95
Kinyarwanda (kin)	76.16 ± 0.30	76.21 ± 0.43	75.16 ± 0.94	74.76 ± 0.35
Nigerian Pidgin (pcm)	70.56 ± 0.51	72.12 ± 0.42	71.81 ± 0.52	68.65 ± 0.77
Wolof (wol)	79.63 ± 0.28	78.06 ± 0.76	76.99 ± 0.42	76.27 ± 0.76
Yoruba (yor)	77.71 ± 0.66	77.75 ± 0.81	76.13 ± 0.42	75.33 ± 0.45
Hausa (hau)	79.66 ± 0.61	80.75 ± 0.63	79.75 ± 1.25	78.55 ± 0.60
Xhosa (xho)	70.38 ± 0.67	66.00 ± 0.80	61.08 ± 1.72	58.20 ± 1.30

(a) Mean Labeled Attachment Score (LAS) $\pm$ standard deviation.

Language	Biaffine LSTM	AfroXLMR-large	RemBERT	Stack-Pointer
French (fra)	95.53 ± 0.08	97.30 ± 0.03	97.17 ± 0.07	94.69 ± 0.08
Romanian (ron)	92.45 ± 0.14	95.38 ± 0.08	95.26 ± 0.04	92.01 ± 0.11
Afrikaans (afr)	88.74 ± 0.31	91.37 ± 0.33	90.42 ± 0.01	88.68 ± 0.59
Swahili (swa)	88.57 ± 0.37	89.39 ± 0.22	87.54 ± 0.57	84.92 ± 0.82
Kinyarwanda (kin)	86.44 ± 0.24	86.59 ± 0.26	86.11 ± 0.54	85.71 ± 0.32
Nigerian Pidgin (pcm)	80.41 ± 0.72	82.13 ± 0.30	81.83 ± 0.44	78.12 ± 0.48
Wolof (wol)	85.79 ± 0.16	84.88 ± 0.76	84.19 ± 0.47	82.83 ± 0.77
Yoruba (yor)	85.46 ± 0.41	85.97 ± 0.67	84.25 ± 0.39	83.69 ± 0.35
Hausa (hau)	92.30 ± 0.49	92.32 ± 0.29	91.81 ± 0.74	91.47 ± 0.74
Xhosa (xho)	81.70 ± 0.52	76.78 ± 1.07	72.78 ± 1.69	69.80 ± 1.44

(b) Mean Unlabeled Attachment Score (UAS) $\pm$ standard deviation.

Table 3: Parsing performance results across five seeds for LAS and UAS.

Language	AfroXLMR	RemBERT	Stk-Ptr
French	-0.354	-0.311	0.169
Romanian	-0.265	-0.257	-0.014
Afrikaans	-0.216	-0.134	-0.004
Swahili	0.049	0.181	0.311
Kinyarwanda	-0.002	0.042	0.059
Nigerian Pidgin	-0.053	-0.042	0.065
Wolof	0.077	0.130	0.165
Yoruba	-0.002	0.071	0.107
Hausa	-0.054	-0.004	0.055
Xhosa	0.148	0.314	0.411

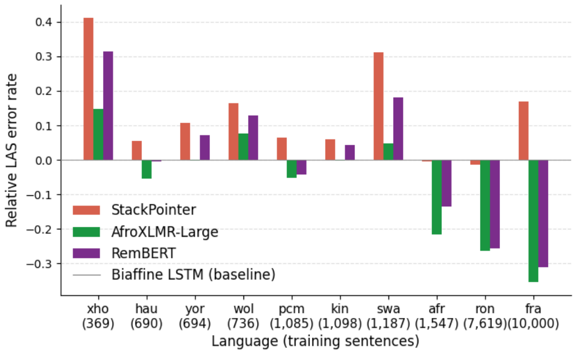
Table 4: RER_{LAS} compared to the Biaffine LSTM baseline. Values $RER_{\text{LAS}} < 0$ favor the comparison model; values $RER_{\text{LAS}} > 0$ favor the LSTM.

Language	AfroXLMR	RemBERT	Stk-Ptr
French	-0.396	-0.367	0.188
Romanian	-0.388	-0.372	0.059
Afrikaans	-0.234	-0.149	0.006
Swahili	-0.072	0.090	0.320
Kinyarwanda	-0.011	0.024	0.054
Nigerian Pidgin	-0.088	-0.072	0.117
Wolof	0.064	0.113	0.208
Yoruba	-0.035	0.083	0.122
Hausa	-0.003	0.064	0.108
Xhosa	0.269	0.487	0.650

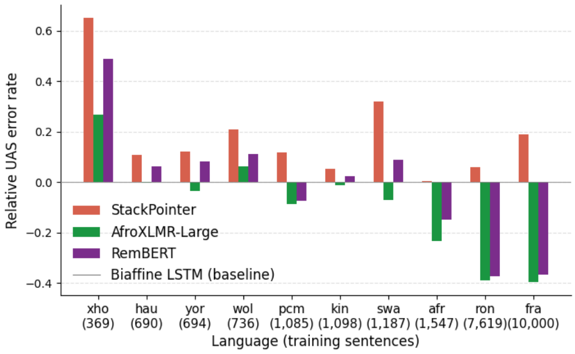
Table 5: RER_{UAS} compared to the Biaffine LSTM baseline. Values $RER_{\text{UAS}} < 0$ favor the comparison model; values $RER_{\text{UAS}} > 0$ favor the LSTM.

For LAS, the slope on `log_train` is $\hat{\beta} = -0.354$ ($p < .001$) for RemBERT and $\hat{\beta} = -0.301$ ($p < .001$) for AfroXLMR-large; the corresponding UAS slopes are steeper at $\hat{\beta} = -0.472$ and $\hat{\beta} = -0.391$ (both $p < .001$). Spearman correlations between training size and mean RER are negative and significant across all model–metric combinations, corroborating this pattern under minimal distributional assumptions. For RemBERT, $\rho_{\text{LAS}} = -0.709$ ($p = .022$) and $\rho_{\text{UAS}} = -0.782$ ($p = .008$); for AfroXLMR-large, $\rho_{\text{LAS}} = -0.745$ ($p = .013$) and $\rho_{\text{UAS}} = -0.891$ ($p < .001$). Fitted curves for LAS are shown in Figure 2.

The fitted curves cross $RER = 0$—the point at which a transformer's predicted error equals the LSTM's—at the sentence counts summarized in Table 6. AfroXLMR-large reaches the crossover at around 830 sentences, while RemBERT requires approximately 1,340–1,390. Several languages in our evaluation fall close to these thresholds: Hausa (690), Yoruba (694), and Wolof (736) sit just below the AfroXLMR-large crossover, while Swahili (1,187), Nigerian Pidgin (1,085), and Kinyarwanda (1,098) fall between the two estimates.



(a) RER_{LAS}



(b) RER_{UAS}

Figure 1: RER vs. sentence count. $RER < 0$ indicates fewer errors than Biaffine LSTM baseline.

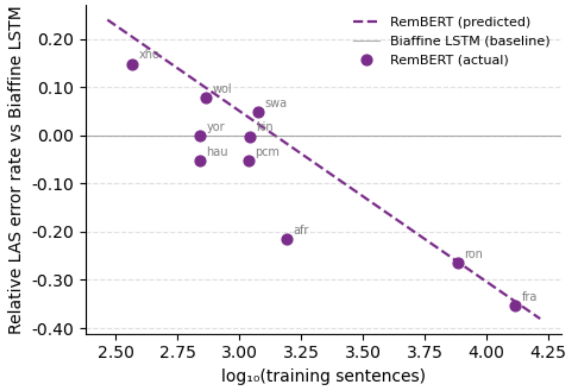
Model	Metric	$\log_{10}$(train)	Sents.
AfroXLMR-large	LAS	2.923	838
RemBERT	LAS	3.142	1,388
AfroXLMR-large	UAS	2.918	829
RemBERT	UAS	3.127	1,339

Table 6: Estimated training-set sizes at which fitted RER curve crosses $RER = 0$, i.e. where transformers' predicted error equals the Biaffine LSTM's.

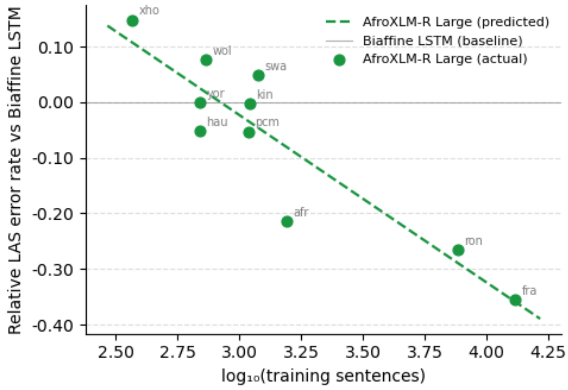
6.3 Morphological Complexity

To assess whether morphological complexity explains residual variance in architecture-level differences beyond what training size captures, we extend the mixed models by adding MATTR as a predictor alongside log training sentences, and compare model fit with and without this term.

For RemBERT, adding MATTR significantly improves fit on both LAS ($\chi^2(1) = 6.01$, $p = .014$) and UAS ($\chi^2(1) = 11.54$, $p < .001$), with positive `mattr_z` coefficients of $\hat{\beta} = 0.063$ ($p = .016$) and $\hat{\beta} = 0.091$ ($p < .001$) respectively: higher morphological complexity is associated with a positive shift in RER, even after controlling for data size. For AfroXLMR-large, the UAS model shows a significant effect ($\chi^2(1) = 8.07$, $p = .005$; $\hat{\beta} = 0.057$, $p = .003$), while the LAS improvement is marginal ($\chi^2(1) = 3.40$, $p = .065$; $\hat{\beta} = 0.038$, $p = .092$), suggesting that AfroXLMR's African-language pre-training partially offsets the elevated RER that complex morphology imposes on transformer models. In all extended models, the negative effect of training size on RER remains significant. Figure 3 shows partial regression plots in which variables have been residualized for training size, isolating the complexity effect.



(a) RER_{LAS}: RemBERT vs. Biaffine LSTM



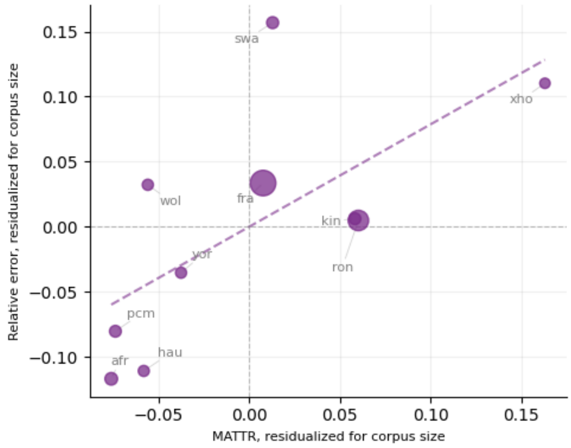
(b) RER_{LAS}: AfroXLMR-large vs. Biaffine LSTM

Figure 2: RER_{LAS} as a function of training size. Plots for RER_{UAS} are omitted for brevity, although the trends are qualitatively identical, with RER_{UAS} decreasing monotonically as training data grows.

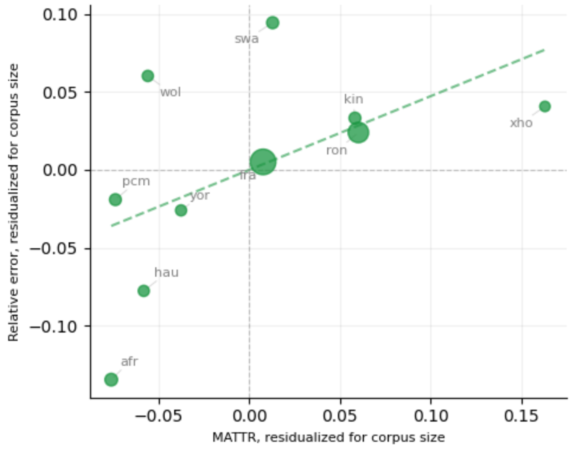
7 Discussion

7.1 Scaling Behavior Across Architectures

Across both transformer models, RER decreases monotonically as training corpus size grows, with transformers achieving $RER \leq 0$ once treebanks reach roughly 800–1,400 training sentences. This pattern is consistent with standard fine-tuning dynamics: multilingual pre-trained models carry useful cross-lingual syntactic representations, but adapting those representations to a specific tree-



(a) RemBERT: RER_{LAS}



(b) AfroXLMR-large: RER_{LAS}

Figure 3: Partial regression of RER_{LAS} on MATTR for RemBERT (top) and AfroXLMR-large (bottom), after residualizing both variables for training size. Partial regression plots for RER_{UAS} are omitted for brevity, although the relationship of RER_{UAS} with morphological complexity is consistent with that of RER_{LAS} and leads to the same conclusions.

bank's annotation conventions requires sufficient supervision. With little training data, fine-tuning is underconstrained—the model lacks sufficient signal to effectively map its pre-trained representations to the specific structural conventions of the target treebank. The Biaffine LSTM, learning its representations entirely from the treebank, avoids this problem; its inductive biases toward locality and sequential composition suit the short-range dependencies that dominate treebank data. As supervision increases, transformers gain the signal needed to resolve this ambiguity, RER crosses zero, and their cross-lingual transfer becomes advantageous.

The estimated crossover points are practically significant: a substantial fraction of African UD treebanks fall within the 838–1,388 sentence range, meaning the LSTM remains the stronger choice for many under-resourced languages in practice. RER_{UAS} shows a steeper scaling trajectory than RER_{LAS}. We attribute this to structural attachment being more typologically general and therefore more transferable via pre-training, while label assignment depends more heavily on language-specific annotation conventions that neither scale nor pre-training fully resolves.

One limitation of this analysis is the distributional gap in our language sample: the largest low-resource treebank (Afrikaans, 1,547 sentences) is separated from the smallest high-resource treebank (Romanian, 7,619 sentences) by nearly a fivefold difference, with no languages in the 2,000–6,000 sentence range. The crossover estimates therefore involve some extrapolation, and a sample with languages in the intermediate range would better characterize the transition region.

7.2 Morphological Complexity and the Low-Resource Penalty

Even after controlling for corpus size, morphological complexity predicts a consistent positive shift in RER for both transformer models. Morphologically rich languages produce a larger inventory of distinct inflected forms, compounding the difficulty of learning an accurate parser from limited data. The effect is weaker for AfroXLMR-large—its MATTR coefficient is roughly 1.5–1.6 times smaller than RemBERT's—which is consistent with its continued pre-training on African-language text providing richer coverage of inflectional paradigms, partially offsetting the morphological component of RER.

In absolute terms the MATTR effect is modest, and interpretation is complicated by Xhosa's status as simultaneously the smallest treebank and most extreme MATTR outlier, giving it disproportionate leverage over the complexity estimates. We also note that MATTR captures lexical diversity rather than morphological complexity directly. More principled operationalizations—such as morpheme-per-word ratios or paradigm entropy—would provide a stronger test of this hypothesis in future work. While lemmatizing the treebanks could theoretically mitigate this morphological penalty, robust lemmatizers for many of these languages remain largely unavailable.

7.3 Stack-Pointer Underperformance

The Stack-Pointer parser shows $RER > 0$ on nearly every language, suggesting that its architectural complexity is counterproductive in this data regime. Its top-down, depth-first decoding introduces error propagation, since mistakes at high nodes cascade to descendants—an effect that is exacerbated when data is scarce. The pointer network decoder also adds substantially more parameters than the Biaffine LSTM, increasing the risk of overfitting on small treebanks, and the higher-order sibling and grandparent features that distinguish this architecture depend on prior parsing decisions that may themselves be unreliable under low-resource conditions.

7.4 Broader Limitations

Several factors limit the generalizability of our conclusions. (1) Our treebanks differ in domain and genre, and some of the observed RER differences may reflect genre mismatch rather than resource level. (2) For non-transformer models, the quality of word embeddings introduces a confound that remains unquantified in our crossover estimates, since higher-resource languages benefit from embeddings trained on larger corpora. (3) Our transformer baselines are all multilingual—languages with monolingual pre-trained encoders available may show different crossover behavior. (4) Typological differences in word order and non-projectivity affect parsing difficulty independently of resource level and cannot be fully separated from it in our sample. (5) The test set for our lowest-resource language contains fewer than 50 sentences, limiting the statistical precision of reported means. A larger and more diverse language sample would provide more robust grounding for our claims.

8 Conclusion

This work examined whether transformer-based parsers maintain their advantage over recurrent architectures when training data is scarce. Across ten typologically diverse languages spanning a wide range of resource levels, our results consistently show $RER > 0$ for both AfroXLMR-large and RemBERT on the lowest-resource languages, while RER decreases monotonically with training data until transformers achieve $RER \leq 0$ at higher resource levels. Statistical analysis confirms a significant negative relationship between training set size and RER, and morphological complexity—operationalized via MATTR—emerges as a secondary predictor of elevated RER: higher morphological richness compounds the transformer's error disadvantage, an effect that is attenuated for AfroXLMR-large, likely due to its pre-training on African-language text. These findings suggest that data scarcity and morphological richness jointly constrain the effectiveness of multilingual transformer fine-tuning in low-resource conditions.

The LAS scores on the lowest-resource languages remain lower than most downstream applications require, but they nonetheless carry practical value. Imperfect parsers may still assist with annotation, reducing the clerical burden on human annotators and accelerating treebank development. Parsing accuracy also functions as a sensitive diagnostic of the overall NLP infrastructure available for a language. Taken together, these results suggest that the Biaffine LSTM may be better suited for syntactic tool development in under-resourced languages—for which $RER > 0$ favors it over transformer-based models—until sufficient annotated data is available to leverage the representational capacity of large pre-trained models.

Future work. Several directions would strengthen and extend these findings. The precise $RER = 0$ crossover could be localized by systematically subsampling larger treebanks to produce controlled, within-language training sets of varying sizes. Whether the crossover generalizes beyond dependency parsing to structurally simpler tasks such as POS tagging and NER, and whether zero-shot or few-shot cross-lingual transfer can push it toward lower resource levels, are open questions. The evaluation should also be extended to morphologically rich agglutinative languages from Southeast Asia and the Americas, which are not represented in the current study, and to architectures such as the TreeCRF parser (Zhang et al., 2020), which has been reported to perform well under low-resource conditions. Finally, more principled measures of morphological complexity—such as information-theoretic metrics—would allow a cleaner test of whether the elevated RER for transformers is specifically driven by inflectional disambiguation load, and evaluating these architectures on lemmatized data, where available, could test if neutralizing inflectional variance mitigates this MATTR confound.

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